SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Term 7- Sept 2025

Nanoelectronics and Technology (01.119/99.503)-Week 4 Class 2

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09-Oct- 2025

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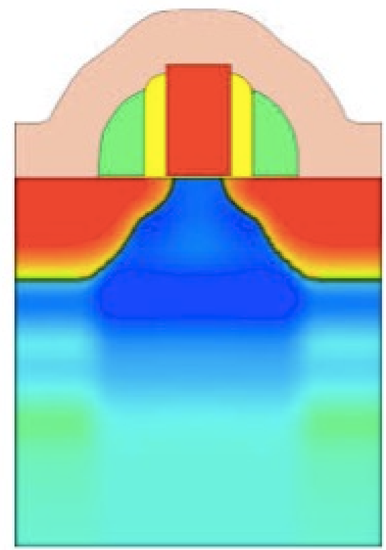
Outline

- Interconnect Scaling and Issues
- Silicon wafer preparation
- IC Cleanroom
- IC Fabrication

FinFET : Challenges

- Expecting higher temperatures for new FinFET Devices
  - Shorter gates and higher fins increase temperatures
  - Heat accumulates due to narrow fin structure and lower thermal connectivity material used
  - Higher channel temperature for higher power consumed
- Design thermal impacts
  - Performance
    - Mobility, threshold voltage, and saturation, etc. all depend on temperature
    - Impact circuit performance
  - Reliability challenges
    - Electromigration (EM), higher temperature degrades lifetime
    - BTI, and HCI can have strong temperature dependencies → high temperature worsens device degradation
- Simulation needs to estimate temperature rise in individual device and surrounding structures



Aging and Self-Heating in FinFETs- Paul McLellan

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FinFET : Summary

Advantages

- **Improved Electrostatic Control**
- **Reduced Leakage Current**
- **Higher Drive Current and scalable**
- **Enhanced Integration with Si Technology**

Challenges

- **Design and Manufacturing Complexity**
- **Variability**
- **Thermal Management**
- **Higher Parasitic Capacitance**

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Interconnect Scaling

Two drivers for Interconnect Scaling:

- RC delay
- Electro-migration
- Stress-migration

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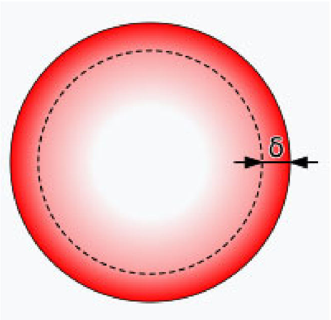
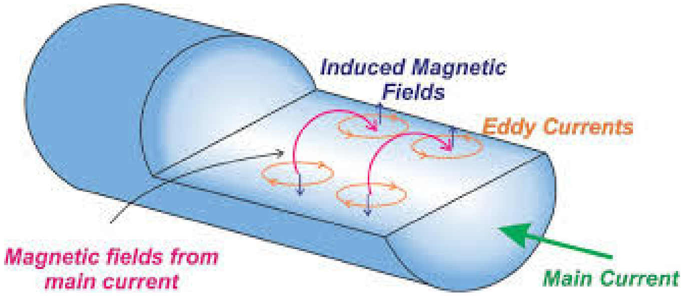
Skin depth:

- In electromagnetism, **skin effect** is the tendency of an alternating current (AC) to become distributed within a conductor such that the current density is largest near the surface of the conductor and decreases exponentially with greater depths in the conductor.



- It is caused by opposing eddy currents induced by the changing magnetic field resulting from the alternating current. The electric current flows mainly at the *skin* of the conductor, between the outer surface and a level called the **skin depth**.



https://en.wikipedia.org/wiki/Skin_effect

https://www.allaboutcircuits.com/technical-articles/understanding-the-skin-effect-cylindrical-and-rectangular-conductors-eddy-currents-and-current-crowding/

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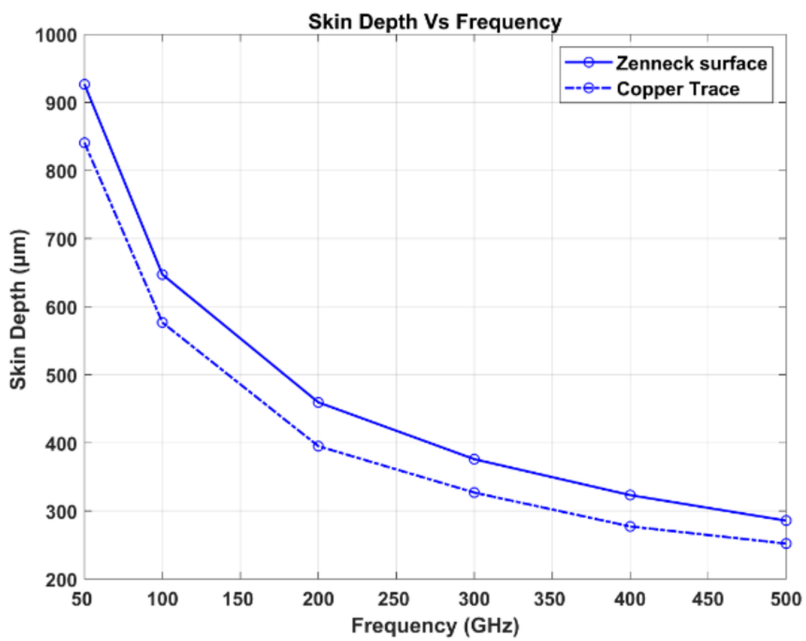

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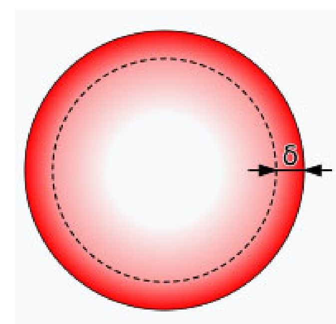

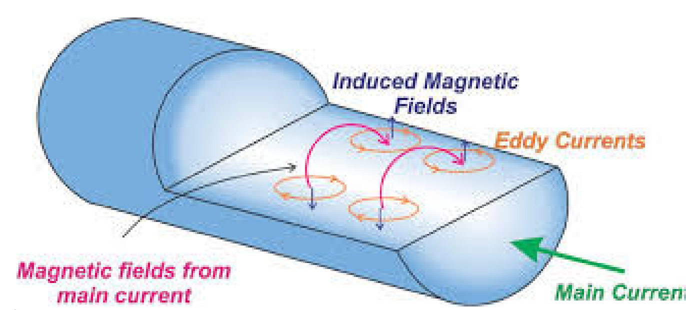

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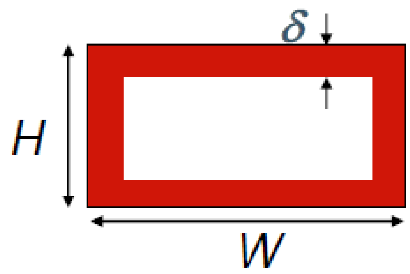
Skin depth:

- At high enough frequencies, R can increase due to ***skin effect***
- Current flow constrained in a thin layer along the surface of the conductor: **R increases**
- **Skin depth**, δ is defined as the depth at which current falls of to a value of 1/e of its nominal value



$$\delta = \sqrt{\frac{\rho}{\pi f \mu}}$$

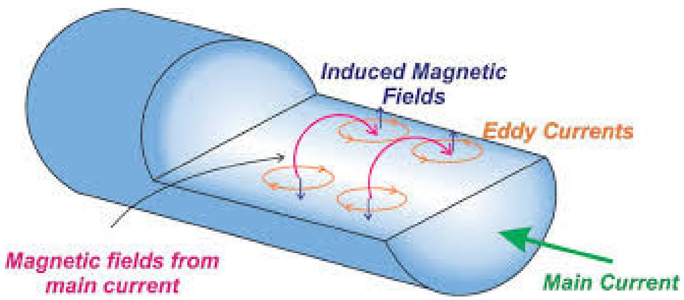
μ is the permeability of the surrounding dielectric

ECE 225 / Kaustav Banerjee

$$\delta = \sqrt{\frac{2\rho}{\mu\omega}}$$

where:

- ρ is the resistivity of the material,
- μ is the permeability,
- $\omega = 2\pi f$ is the angular frequency.



- For Cu wires:

Freq	1 GHZ	5 GHZ	10 GHZ	15 GHZ	20 GHZ
Skin depth (μm)	2.08	0.93	0.66	0.53	0.46

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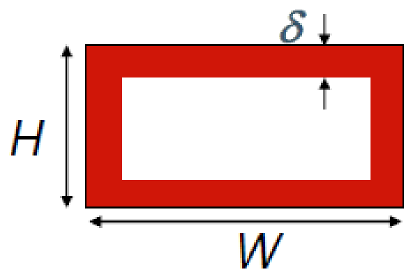

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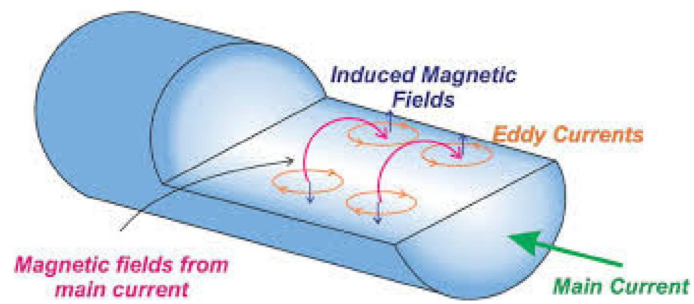

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Skin depth:

Design Considerations: In high-frequency applications (like RF circuits or high-speed digital circuits), designers often take skin depth into account to optimize interconnect design.

In high-frequency scenarios, as it directly impacts performance, efficiency, and reliability.

For example:

- **Larger Conductors**: Using wider wires can help mitigate the increased resistance due to skin effect at high frequencies.

- **Plating Techniques**: Techniques like electroplating can be used to create a conductive layer on a substrate that maximizes the effective current-carrying area.

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Interconnect scaling- Stress Migration

- **Stress Migration** is a reliability failure mechanism observed primarily in **metal interconnects** of CMOS integrated circuits. It occurs due to **mechanical stress gradients** within the metal lines, even *without any electric current* (unlike electromigration).

- Thermal-mechanical stresses cause vacancies to accumulate and form voids in metal lines and vias, potentially leading to circuit failure.

- These stresses arise from the difference in thermal expansion coefficients (CTE) between the metal and dielectric materials.

- Stress migration is driven by stress gradients, leading to atomic mass transport and void formation, especially at locations with concentrated tensile stress, such as at via corners.

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Interconnect scaling- Stress Migration

Physical Origin

- During **deposition and cooling** of metal interconnects (like aluminum or copper), **thermal expansion mismatch** occurs between the metal and surrounding dielectric (e.g., SiO_2 or low-k material).

- As temperature decreases, **compressive stress** builds up in the metal.

- At elevated operating temperatures or during thermal cycling, **stress relaxation** takes place by atomic diffusion — atoms migrate from regions of **compressive stress** to **tensile stress** regions.

- This leads to **void formation** (depletion of atoms) or **hillock formation** (accumulation of atoms).

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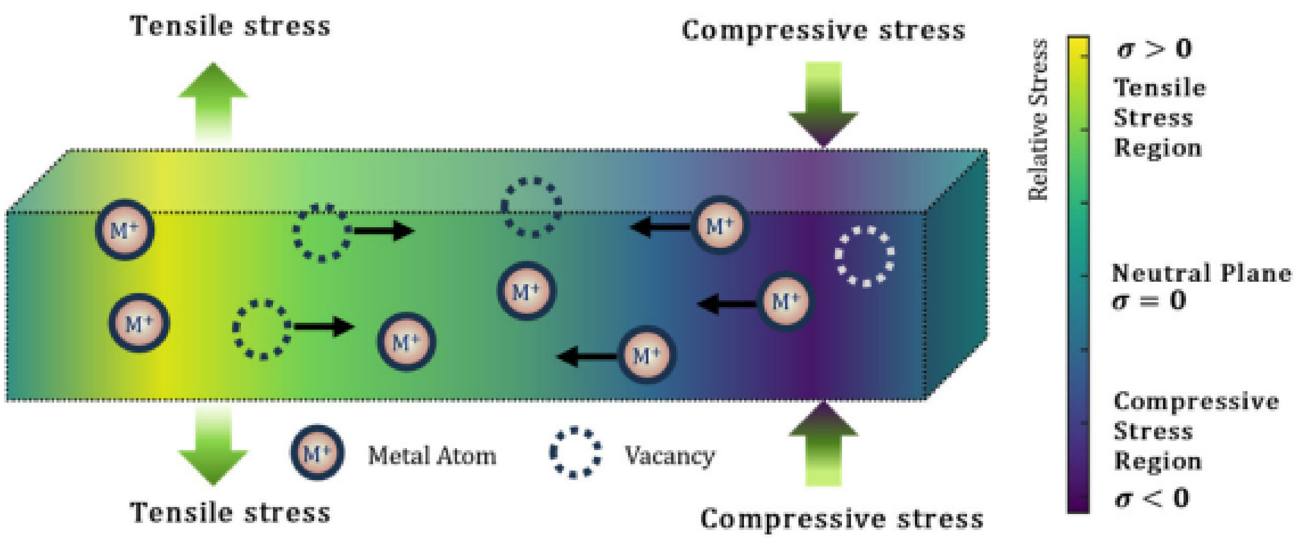
Interconnect scaling- Stress Migration

Thermal mismatch Different coefficients of thermal expansion (CTE) between Cu/Al and surrounding dielectric (SiO_2, SiN, etc.). During cool-down from processing (~400–450 °C) to ambient (~25 °C), Cu contracts more → develops **tensile stress**.



Schematic illustration of SM, depicting the movement of atoms driven by a stress gradient within the material. https://doi.org/10.3390/electronics14153151

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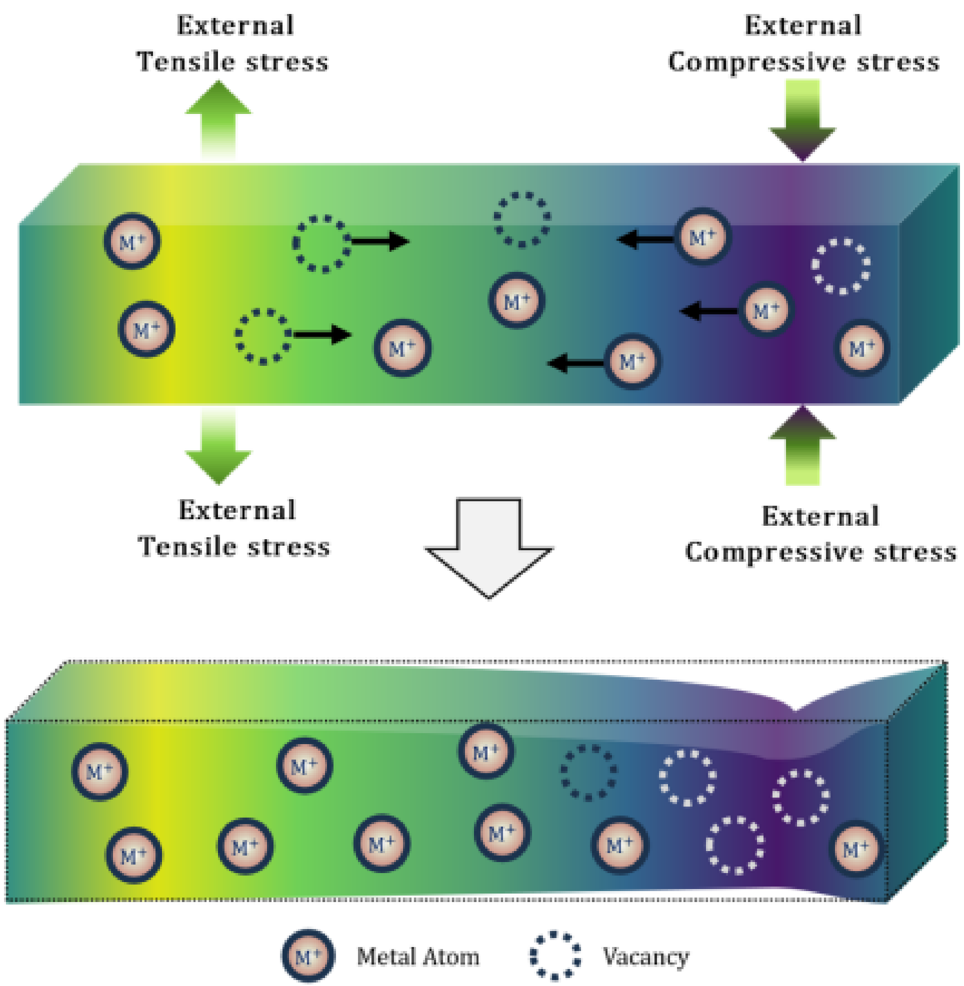
Interconnect scaling- Stress Migration



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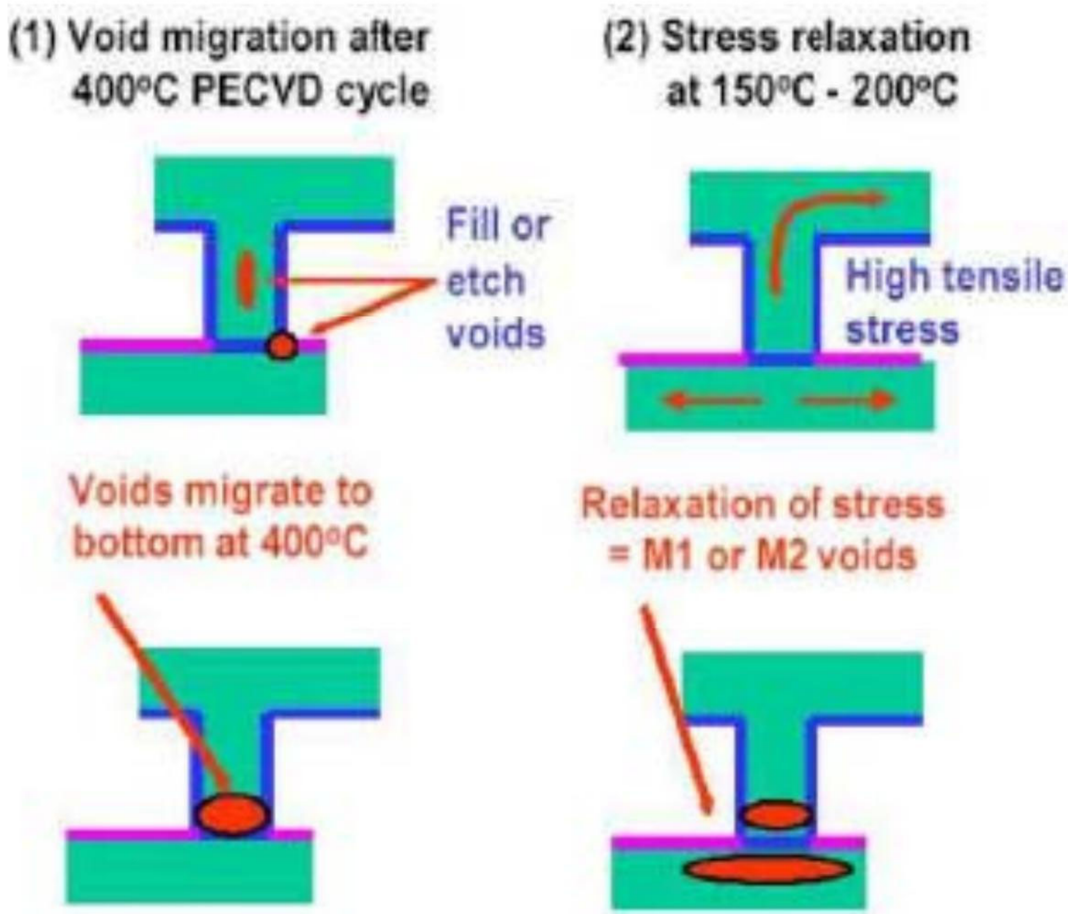
Interconnect scaling- Stress Migration



https://www.academia.edu/53952750/Effect_of_stress_migration_on_electromigration_for_nano_scale_advanced_interconnects

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Interconnect scaling- Stress Migration

Aspect	Description
Driving force	Stress gradient ($\partial\sigma/\partial x$) — *not* current density
Dominant diffusion path	Grain boundary or interface diffusion
Temperature dependence	Exponential (Arrhenius) via diffusion coefficient
Typical locations	Near vias, line ends, or metal–dielectric interfaces

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Interconnect scaling- Stress Migration

Difference from Electromigration (EM)

Feature	Stress Migration	Electromigration
Driving force	Stress gradient	Electron wind force
Current	Not required	Required
Direction	From compressive → tensile	Along electron flow
Temperature sensitivity	Moderate	High
Failure mode	Void or hillock	Open or short circuit

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Interconnect scaling- Stress Migration

Mitigation Techniques

- Use of **Cu interconnects** with proper barrier layers (Ta/TaN).
- **Annealing** to reduce residual stress.
- **Capping layers** (e.g., SiN, CoWP) to control diffusion.
- **Optimized layout** (shorter line lengths, redundant vias).
- **Use of low-k dielectrics** to reduce thermal mismatch.

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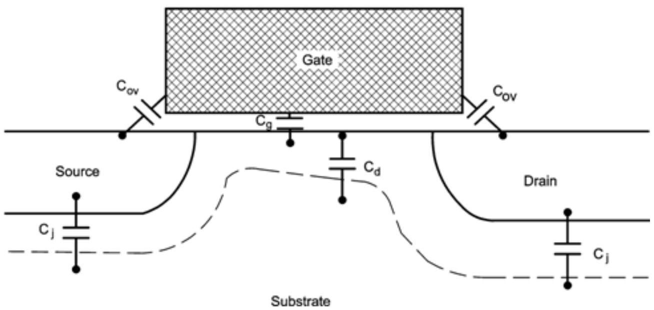
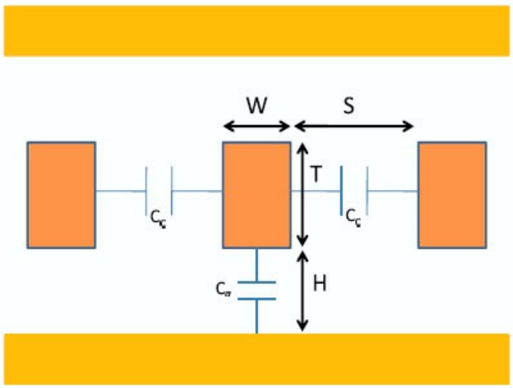
Interconnect scaling

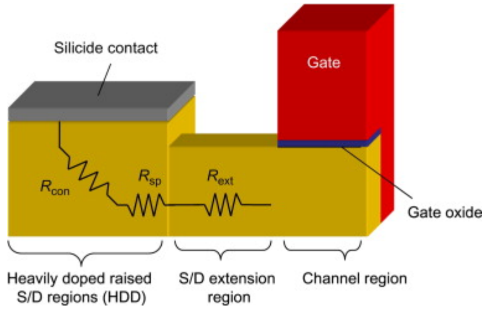
Parasitic Capacitances and Resistances

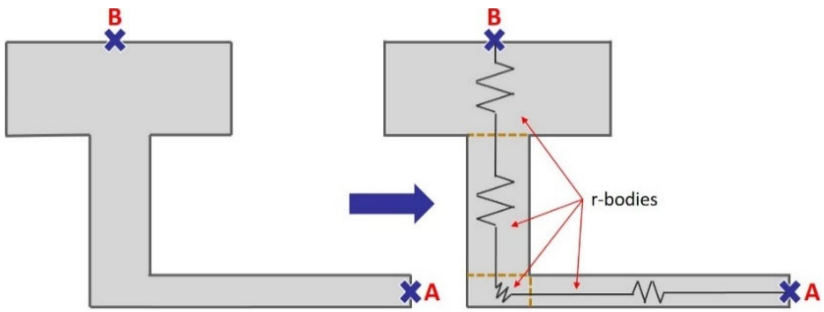
Interconnects



MOSFET parasitic capacitances







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Interconnect Scaling

Requirements of the interconnection materials

- Low resistivity of conductors
- Low capacitance => low dielectric constant
  - Low RC delay
  - Low cross talk
  - Low power dissipation (CV^2f loss)

Ref: Prof. Saraswat, EE 311 Notes on interconnects, Stanford University

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Interconnect scaling

- Resistance to electromigration
- Ease of deposition of thin films of the material
- Ability to withstand the chemicals and high temperatures required in the fabrication process
- Good adhesion to other layers - low physical stress
- Stability of electrical contacts to other layers
- Ability to contact shallow junctions and provide low resistance
- Ability to be defined into fine patterns

Ref: Prof. Saraswat, EE 311 Notes on interconnects, Stanford University

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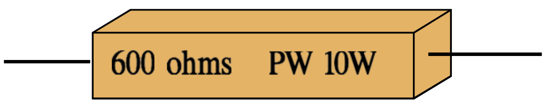
Resistors on ICs

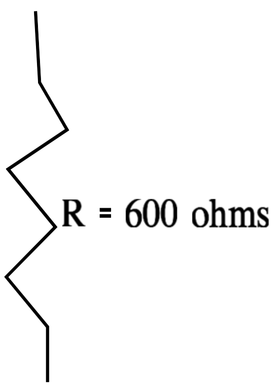
Resistors are fabricated on ICs using

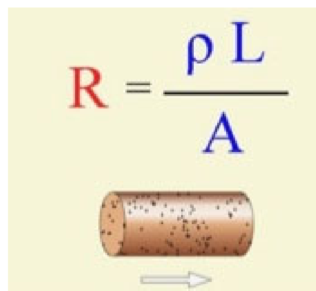
- either the diffused resistor method- doping specific regions of the semiconductor material,
- or the thin-film resistor method, where a resistive thin film is deposited and patterned on the substrate

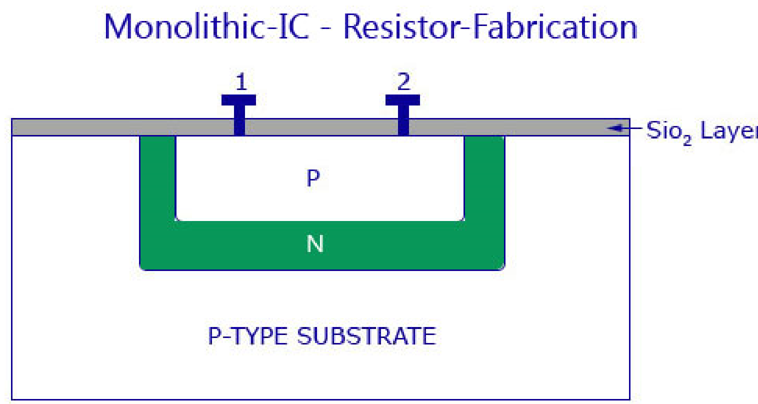
A)



B)

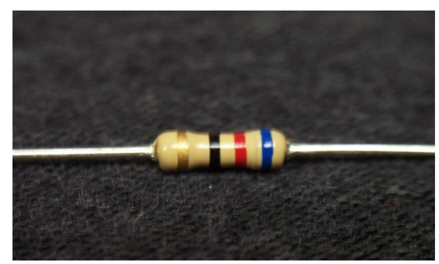






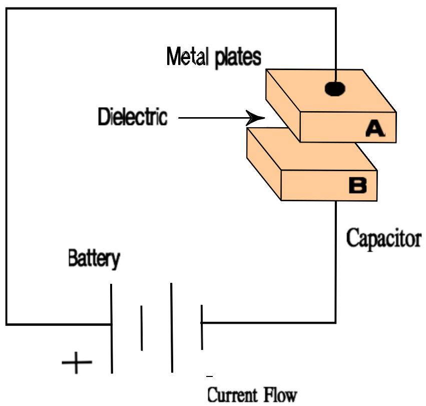
🗝 Key Theme:

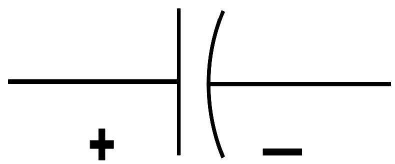
- Resistance is measured in **ohms**



Capacitors on ICs

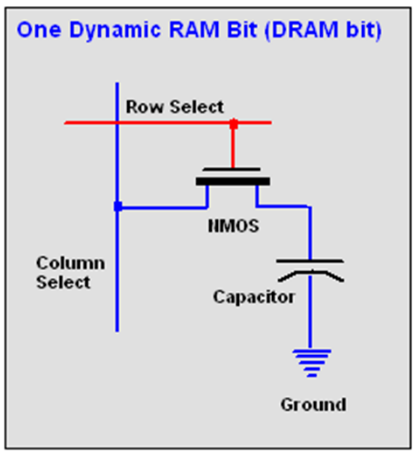
- Capacitance is measured in **farads** and contains two conductive plates and a dielectric between the two plates

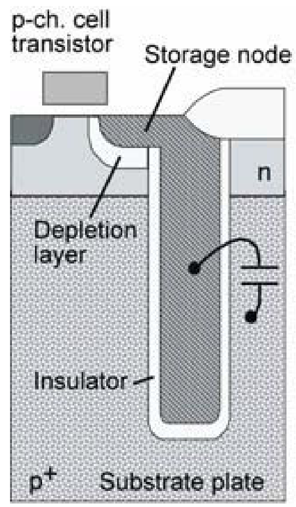




Symbol for a capacitor

Capacitor in DRAM





https://www.intechopen.com/chapters/9855

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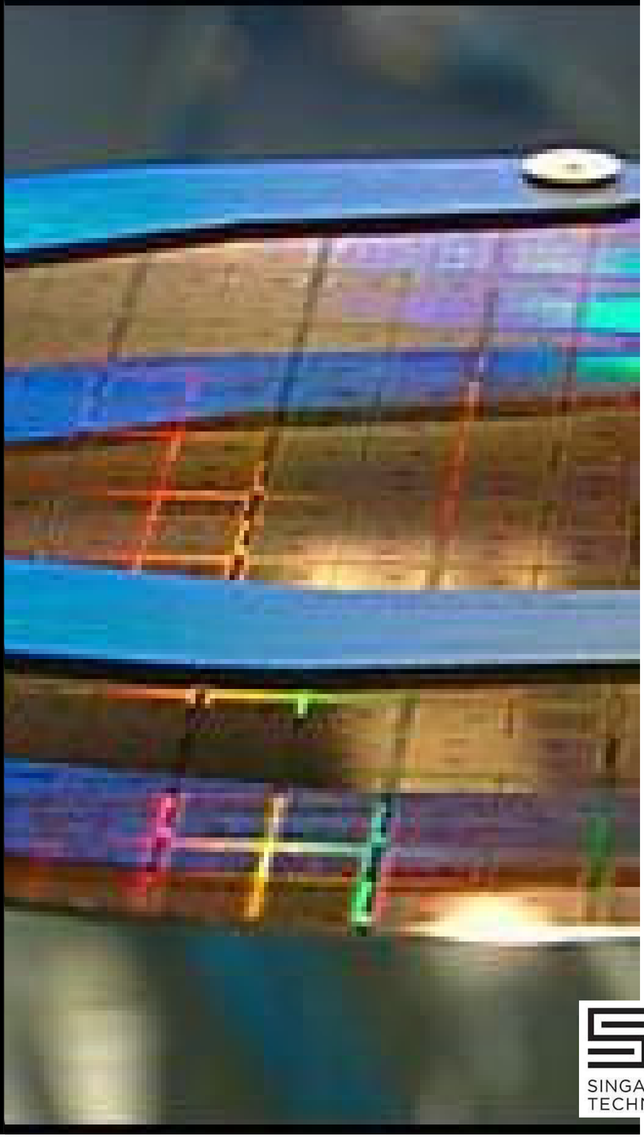
IC Design Approach (Before Fabrication)



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Design and Mask Generation

Objectives:

- Discuss circuit design
- Discuss layout design
- Describe tapeout and GDSII files
- To discuss generation of Reticles and Photomasks
- To discuss IP's

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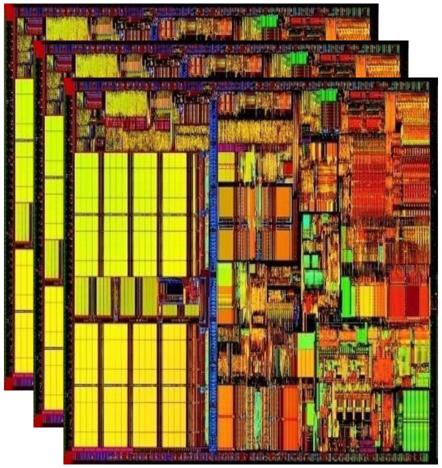
Concept and Specifications Stage

- What technology to use to design the part
- Power supply needs
- Kind of package
- Pin count
- Maximum operating frequency
- Number of gates needed
- Number of I/O pads needed
- Number of buffer gates needed



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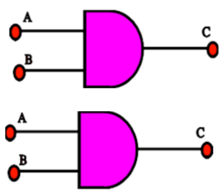
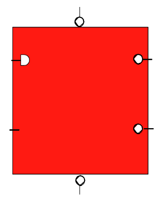
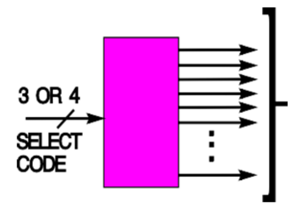
Typical Standard Cells



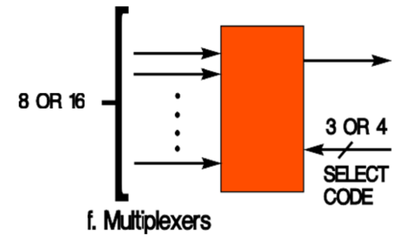
a. Simple gate



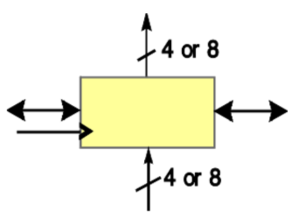
a. Flip-flop



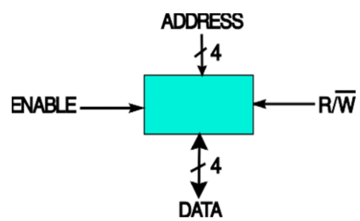
e. Decoder



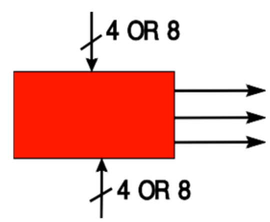
f. Multiplexer



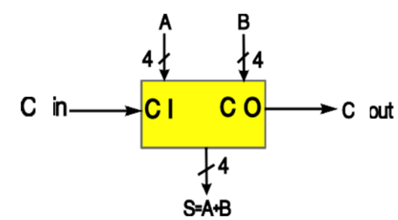
c. Shift Registers and counters



d. RAM and ROMs

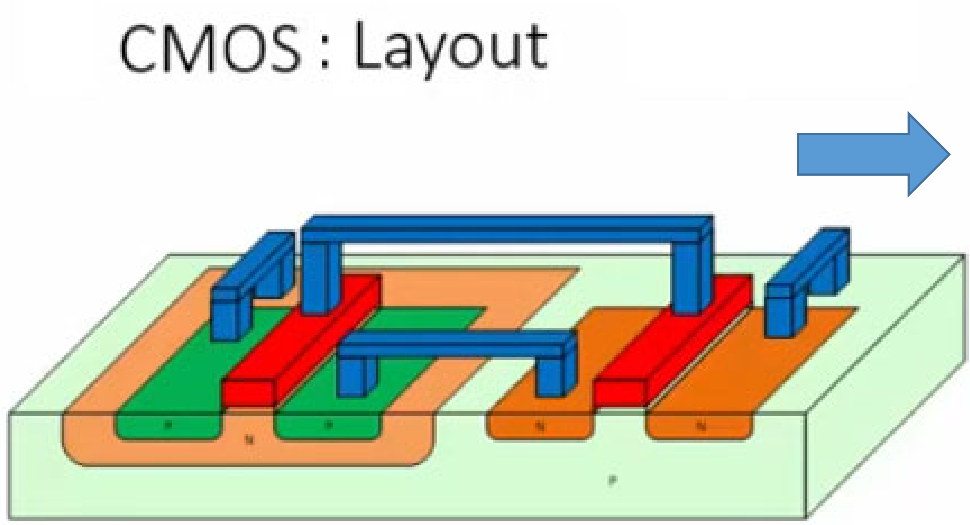


g. Comparator

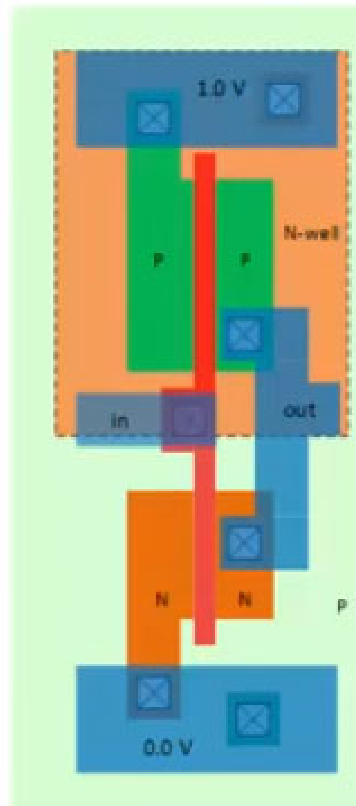


h. Adder

Physical Layout



Side-view



Top-view

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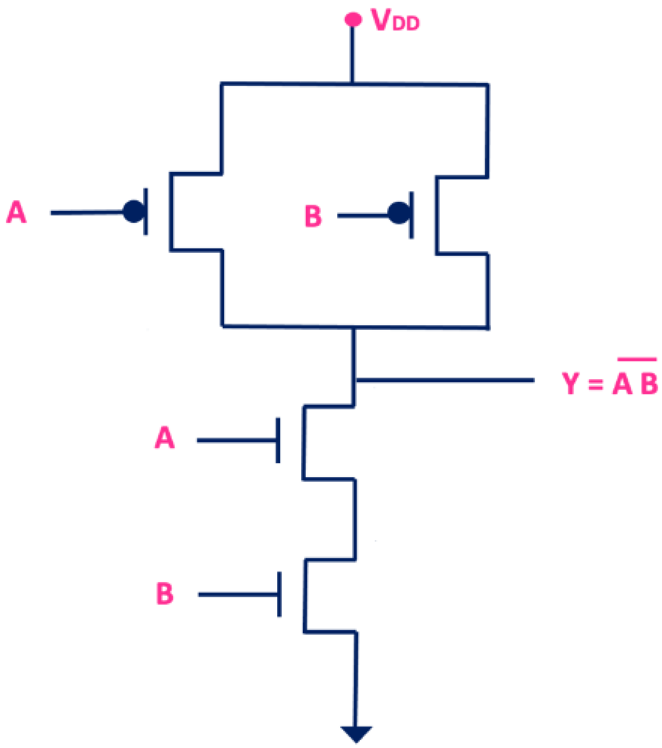
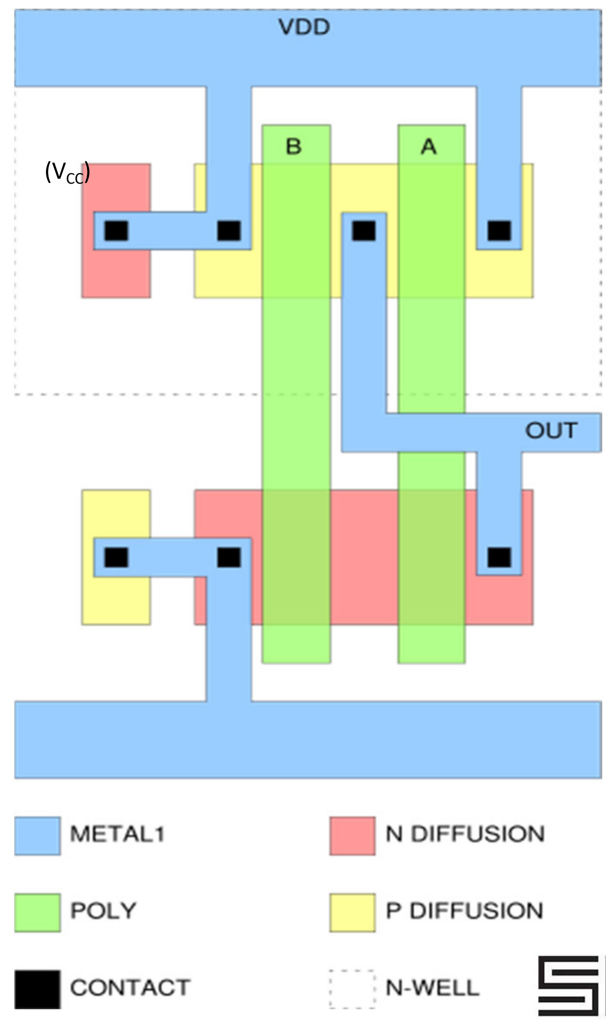
NAND Circuit Layout

Physical layout of NAND circuit from top down view.



Truth Table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



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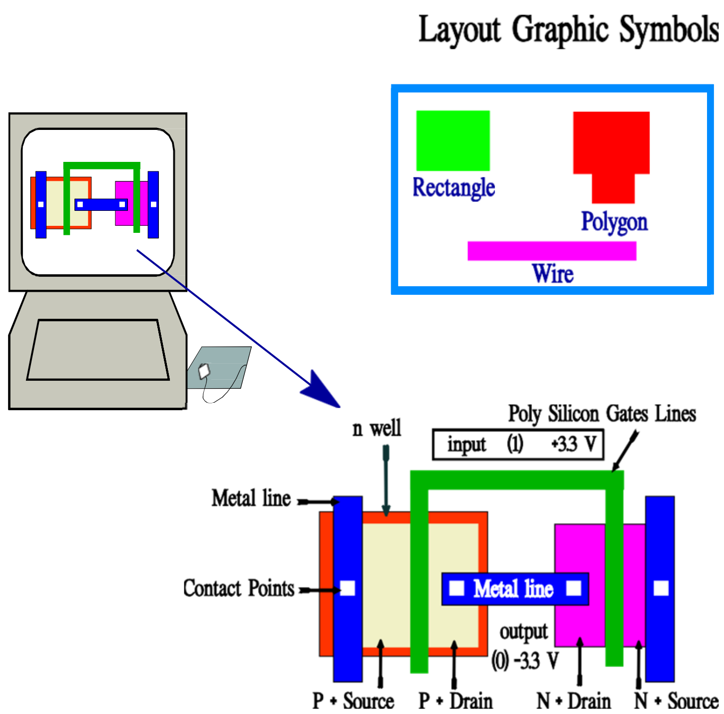
Physical Layout

Physical layout is a process of transforming the electrical schematic into polygons that will be patterned on to the wafer via masks

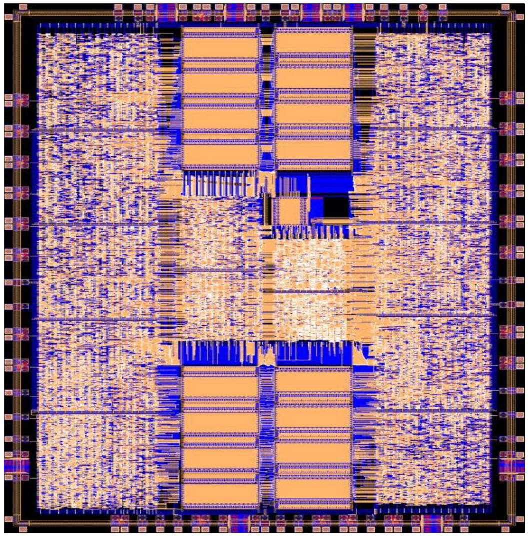


Key Theme:

Could have up to 60 separate layouts per mask set

Electronic Design Aides “EDA” Applications

- To design a complex chip or printed circuit board
- To verify the functionality before creation of the device
- To experiment with different chip designs and package types to observe how performance and cost are impacted
- To save time by entering electronic textual descriptions and to allow the EDA software to generate the electronic circuitry
- To verify and insure that signals are not delayed
- To automate the physical design



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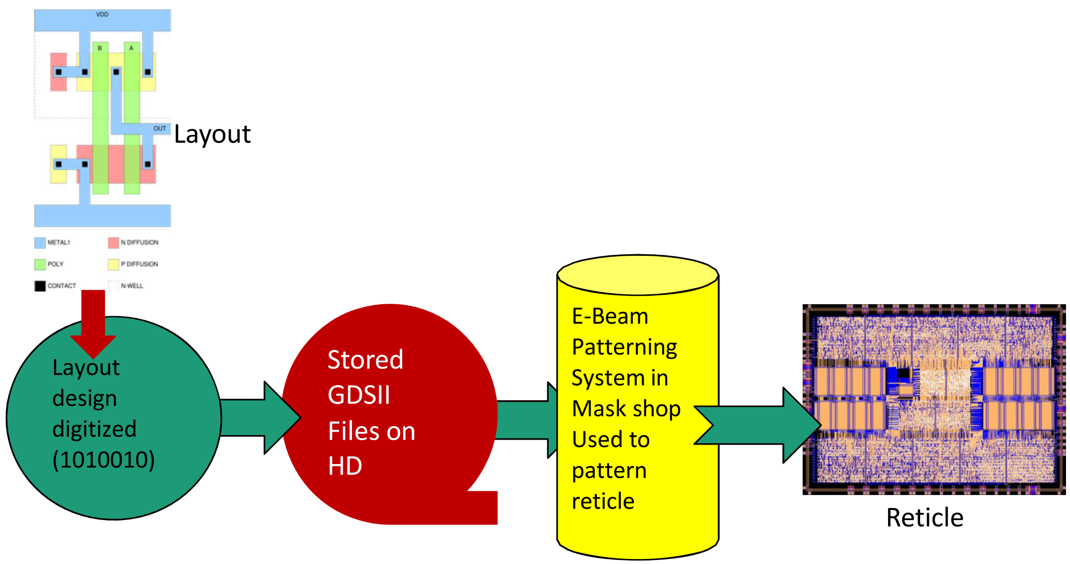
Tape out

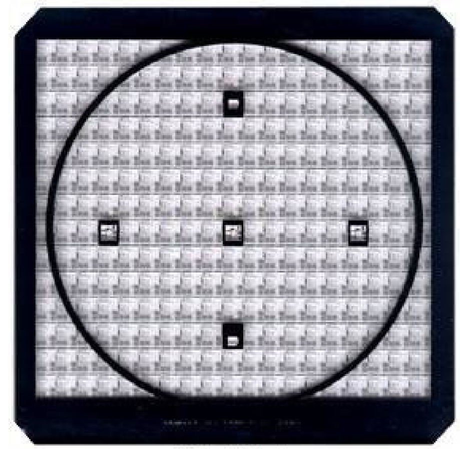
- **Tape out** is a term used in the design area to described the function of digitizing polygons after layout and transferring them to a storage device that will be used to control the patterning of a photomask in the mask shop.
- The files stored on the storage media are referred to as **GDSII files**.



IC Masks

- A **semiconductor IC mask** (also called a **photomask** or **reticle**) is a **master template** that contains the precise **geometric pattern** of one layer of an integrated circuit.

- During photolithography, **UV light** passes through the mask to transfer its pattern onto a **photoresist-coated wafer**, defining where material will later be **etched, doped, or deposited**.



Reticle Mask

- Each step in the fabrication process uses a different mask. A wafer could have up to 60 layers. Masks can be repaired by using a diamond probe to repair opaque defects

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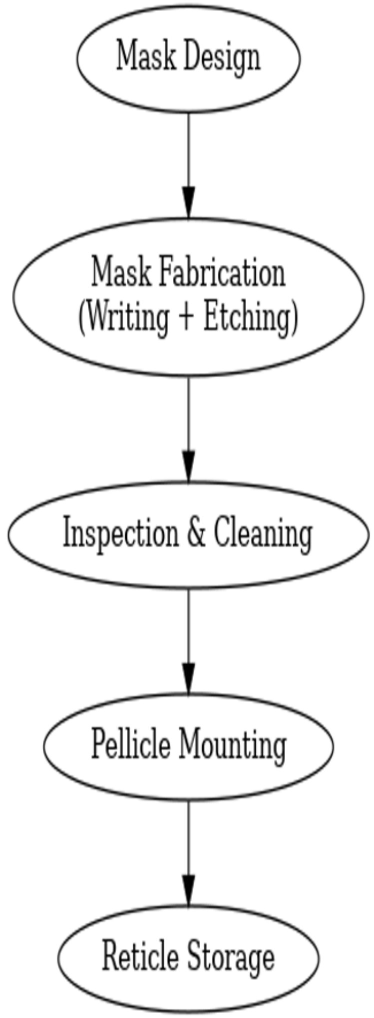
Mask and its Parameters

- Quartz material of choice for steppers.
  - very low coefficient of thermal expansion
  - High light transmission in UV and DUV wavelengths.
- Plate must be free from external and internal defects.
  - Both surfaces must be optically flat.
  - No internal flaws such as air pockets or foreign material (FM) allowed.
- Plates are polished and cleaned
- Quartz plates are coated with UV opaque material.
  - Cr at ~1000Å material of choice via sputtering.
  - Anti-reflective coating applied, Cr_2O_3.

Mask Manufacturing

- Pattern etched into Cr film.
  - resist applied
  - pattern through e-beam or laser methods.
  - resist developed
  - wet etchant used to remove Cr in opening in resist.



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Pellicles

- An IC mask pellicle is a thin, transparent membrane placed on a photomask (reticle) to protect it from dust and other contaminants during the integrated circuit (IC) fabrication process.

- By preventing particles from landing on the mask, pellicles help ensure that printed patterns are free from defects, which is crucial for high-yield production of complex chips, especially during photolithography

- Mounted on frames attached to Cr side of mask. Sometimes mounted to the quartz side as well.

- Material must be thin, transparent to UV, and stable.

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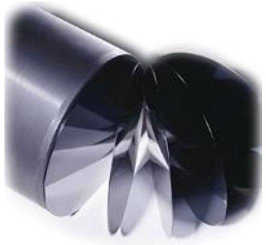
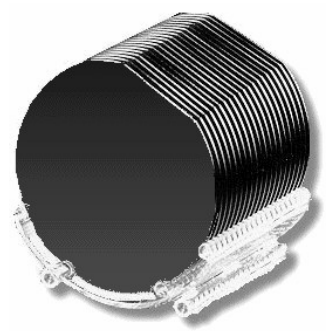
Silicon Crystal Growth and Wafer Preparation

Objectives:

- Define poly silicon and single crystal silicon
- Explain crystal orientation
- Explain how a silicon ingot is formed
- Identify different wafer types
- Explain wafer preparation





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Materials for Integrated Circuits

- First semiconductor transistors built on germanium.
- GaAs, Si, Ge are examples of semiconductor materials.
- Si used in majority of circuits today.
  - Si most abundant element in earth's crust. (easy to obtain, cheap)
  - Si is a elemental semiconductor, compound semiconductors have processing constraints. (temperature, substrate size)
  - With wider band gap, Si functions at higher temperatures than Ge.
  - Si has a stable oxide (SiO_2) that is an insulator and good diffusion barrier.

Dopant	Type
Arsenic (As)	n Type
Phosphorus (P)	n Type
Antimony (Sb)	n Type
Boron (B)	p Type

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Silicon Wafer Preparation

- **Si is purified from SiO_2 (sand) by refining, distillation and CVD.**
- **Bulk silicon is first processed in poly-crystalline form**
- **Crystal growth process used to obtain single-crystal form**
- **It contains < 1 ppb impurities. Pulled crystals contain O ($\sim 10^{18}$ cm^{-3}) and C ($\sim 10^{16}$ cm^{-3}), plus dopants placed in the melt.**

First produce highly pure, poly-crystalline form

Poly Silicon

https://www.physics.muni.cz/~jancely/teamCMV/Texty/RuzneTexty/Krystaly/CrystalStructure&Growth.pdf

Source: USNA EE 452 Course on IC Technology

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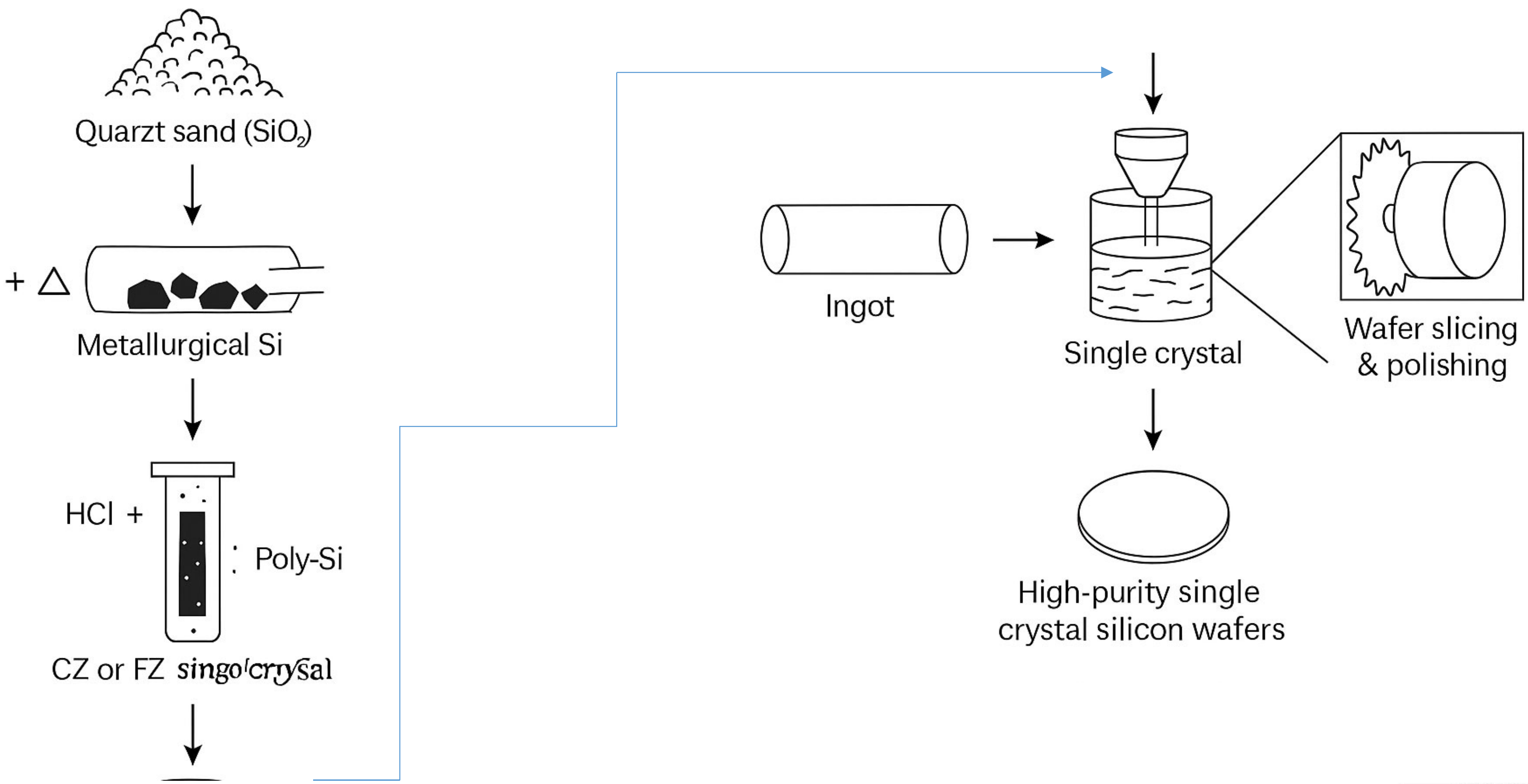
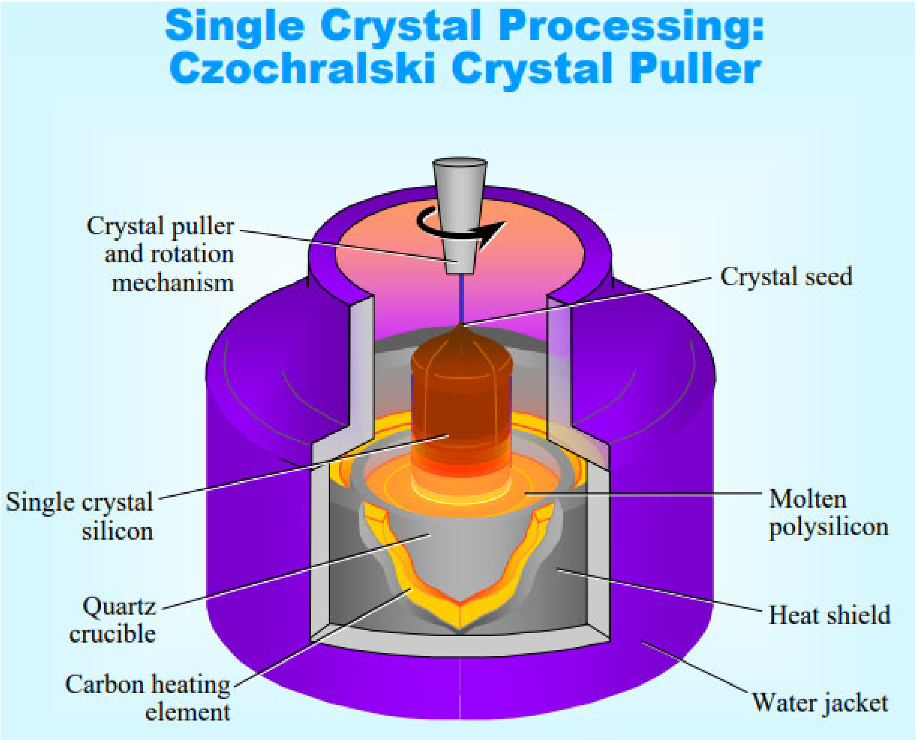
Silicon Wafer Preparation



Silicon Wafer Preparation

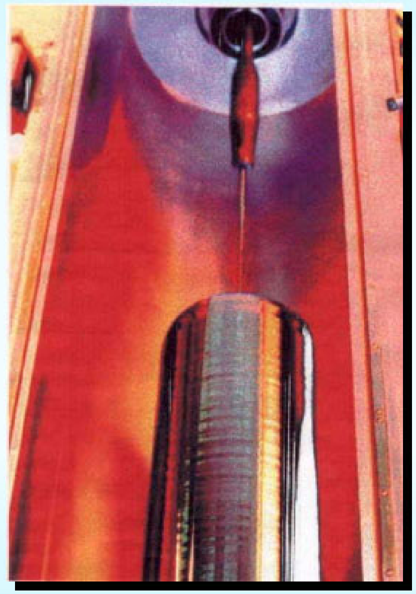


Source: USNA EE 452 Course on IC Technology

https://www.physics.muni.cz/~jancely/teamCMV/Texty/RuzneTexty/Krystaly/CrystalStructure&Growth.pdf

Silicon Wafer Preparation

Silicon Ingot Grown by CZ Method



Photograph courtesy of Kayex Corp., 300 mm Si ingot

Source: USNA EE 452 Course on IC Technology

https://www.physics.muni.cz/~jancely/teamCMV/Texty/RuzneTexty/Krystaly/CrystalStructure&Growth.pdf

Silicon Wafer Preparation

Dopant Concentration

Common Dopants: Phosphorus, Boron, Arsenic

		Concentration (Atoms/cm^3)			
Dopant	**Material Type**	**$< 10^{14}$** (Very Lightly Doped)	**10^{14} to 10^{16}** (Lightly Doped)	**10^{16} to 10^{19}** (Doped)	**$>10^{19}$** (Heavily Doped)
Pentavalent	**n**	$\mathbf{n}^{--}$	$\mathbf{n}^{-}$	**n**	$\mathbf{n}^{+}$
Trivalent	**p**	$\mathbf{p}^{--}$	$\mathbf{p}^{-}$	**p**	$\mathbf{p}^{+}$

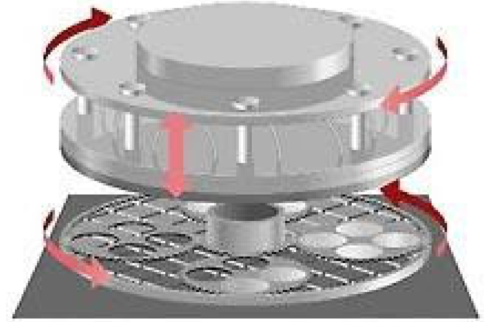
Source: USNA EE 452 Course on IC Technology

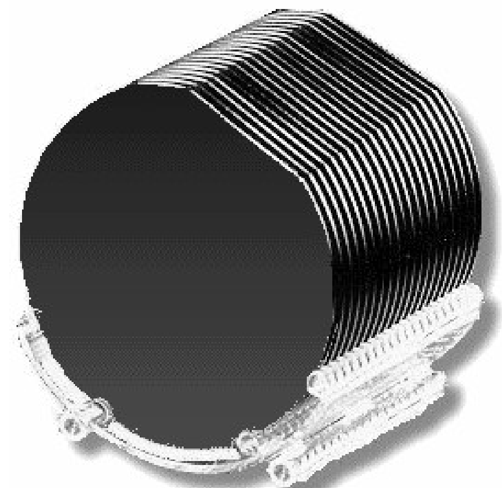
https://www.physics.muni.cz/~jancely/teamCMV/Texty/RuzneTexty/Krystaly/CrystalStructure&Growth.pdf

Silicon Wafer Preparation

Silicon Ingots

Source: USNA EE 452 Course on IC Technology





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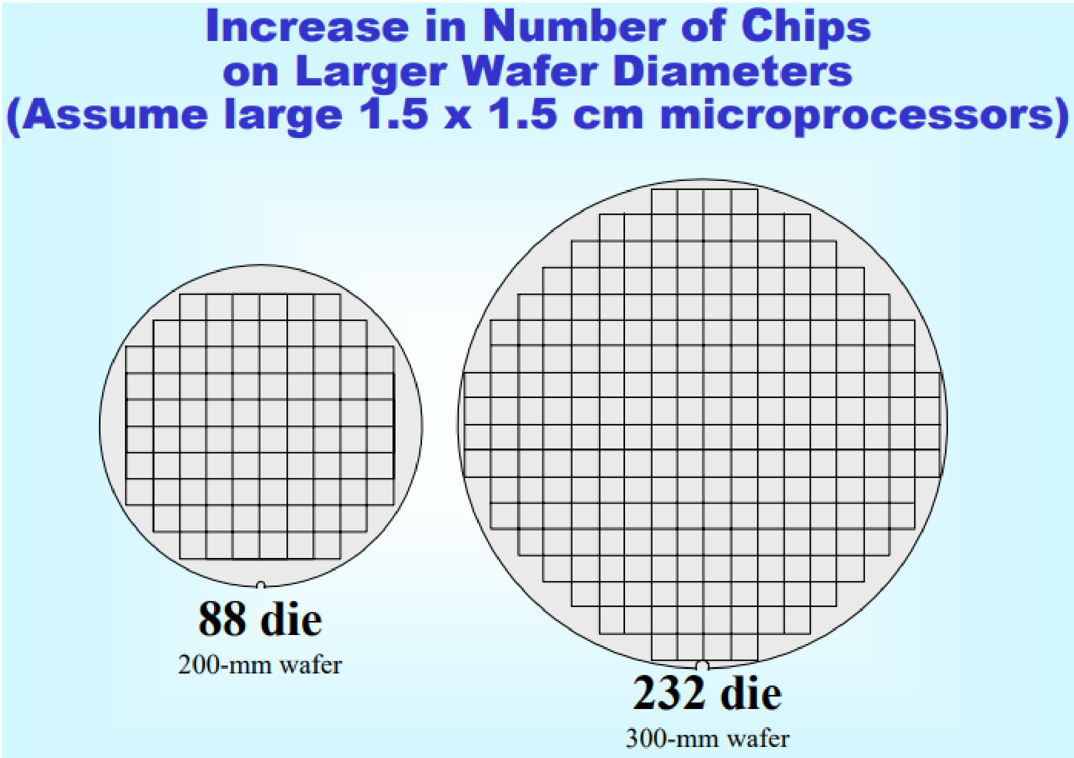
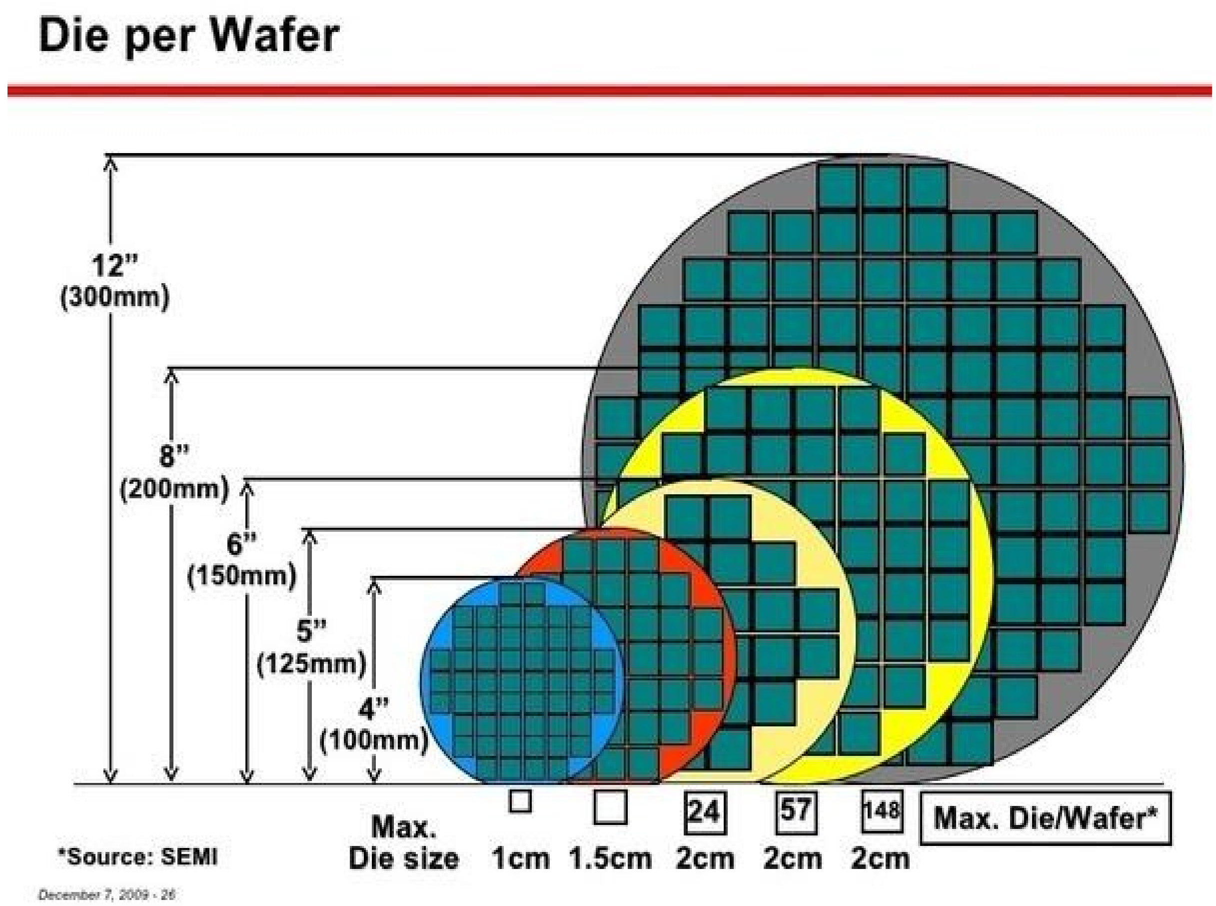
Silicon Wafer Preparation



Source: USNA EE 452 Course on IC Technology

https://www.physics.muni.cz/~jancely/teamCMV/Texty/RuzneTexty/Krystaly/CrystalStructure&Growth.pdf

Wafer Sizes and Die per Wafer



Who makes wafers?

Shin Etsu

Sumco Wacker

Global Wafers

MEMC

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Foundries and Fabless Companies

Top Foundries:
TSMC (Taiwan)
Samsung (South Korea)
Global Foundries (USA)
UMC (Taiwan)
SMIC (China)

Foundries
Produce ICs for other companies

Fabless
Companies have no fabs and use foundries to produce their designed parts

Top Fabless Companies 2nd Qt 2023
Nvidia (USA)
Qualcomm (USA)
AMD (USA)

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450mm Fabs-300mm Fabs

Intel Corp. (Santa Clara, Calif.), Samsung Electronics Co. (Seoul, South Korea), and Taiwan Semiconductor Manufacturing Co. Ltd. (TSMC, Hsinchu, Taiwan) — each to build 22 nm pilot lines at their respective facilities in 2012.

Most likely no production of 450mm wafers in the near future.

<u>**Current Status**</u>

- **Intel, TSMC, Samsung** all explored 450 mm around 2010–2015 (Intel even built pilot lines).
- In 2015–2017, they **shelved 450 mm projects indefinitely**.
- SEMI (industry consortium) officially **froze 450 mm standards development**.
- As of 2025 → **300 mm is still the standard**; instead, cost reduction comes from:
  - **Node scaling** (5 nm, 3 nm, 2 nm).
  - **EUV adoption**.
  - **3D packaging & chiplets**.

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Cleanroom protocol

- Filter the air
- Flow clean air over the product at a low velocity (less than 1 mile per hour)
- Change the entire air volume up to 600 times per hour
- Control personnel practices

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Cleanroom

- **IC cleanroom** is the **heart of semiconductor manufacturing** — the environment where wafers are processed into integrated circuits.

- Cleanrooms are many times cleaner than a medical operating room

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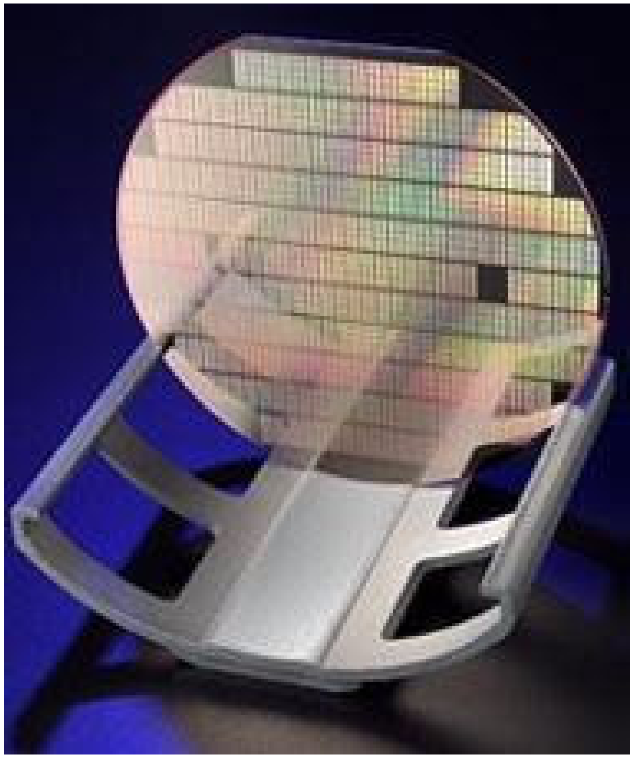
Source of Contaminants in a Cleanroom

Particles:

- Airborne Particulates
- Human Hair and Skin
- Clothing
- Human Breath
- Water
- Gases
- Equipment



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Contamination Sources

- **Facility:** Contamination can come from anywhere in the facility: dirt, debris, dust, shavings, floors, walls, and ceilings. Particles can come from paint (and other coatings) and construction materials like sheetrock or wood.

- **Cleanroom equipment and supplies:** Contamination can also come from clean room equipment and supplies. Friction and wear from equipment use can result in particles released as well as lubricants and emissions from motorized equipment can create contaminates. Vibrating equipment could give off particles.

- **Staff:** Skin shedding accounts for a large proportion of particulates released into a given environment. Oils from skin, hair, and spittle can also contaminate a cleanroom environment. Cosmetics and perfume should be avoided when working in the cleanroom because of their potential contamination risk.

- **Environment and fluids:** Contamination can come from various environmental effects and substances such as fluids. Particles floating in the air can reach cleanroom environments. Microorganisms such as viruses, bacteria, and microbes can also be a source of contamination. Finally, contamination can be generated from the products resulting from the applications. For example, silicon chips can shed particles, and contamination can also come from quartz flakes, aluminum, and other cleanroom debris.

Cleanroom-Standards

Class	Max. particles ≥0.5 μm per ft^3
Class 1	1 particle
Class 10	10 particles
Class 100	100 particles
Class 1000	1,000 particles

https://esg.tsmc.com/en-US/articles/17

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Cleanroom-Ultra-Pure Chemicals

- Many chemicals are used in the fabrication of semiconductor devices. Most chemicals used in semiconductor manufacturing are either liquids or gases.
- Gases are available with very high purity(usually 99.999999 percent)
- Impurities in liquids are harder to control. Since gases are easier to purified
- The trend is to use gases than liquids for chemical reactants

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Clean room

Environmental Push:

Consumption in one average fab:

- 300 tons hazardous chemicals
- 53 liters of water/300mm wafer lot in wet bench alone

  ~ 240 million gallons water/year!
- Today 1.5 KW/wafer

  > 5 GWh electricity/year!

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CMOS Fabrication Steps-CLEANROOM

https://turbofuture.com/industrial/The-World-of-the-Semiconductor-Cleanroom

Video Link:
https://www.youtube.com/watch?v=BIFKDXXPuU4&t=12s

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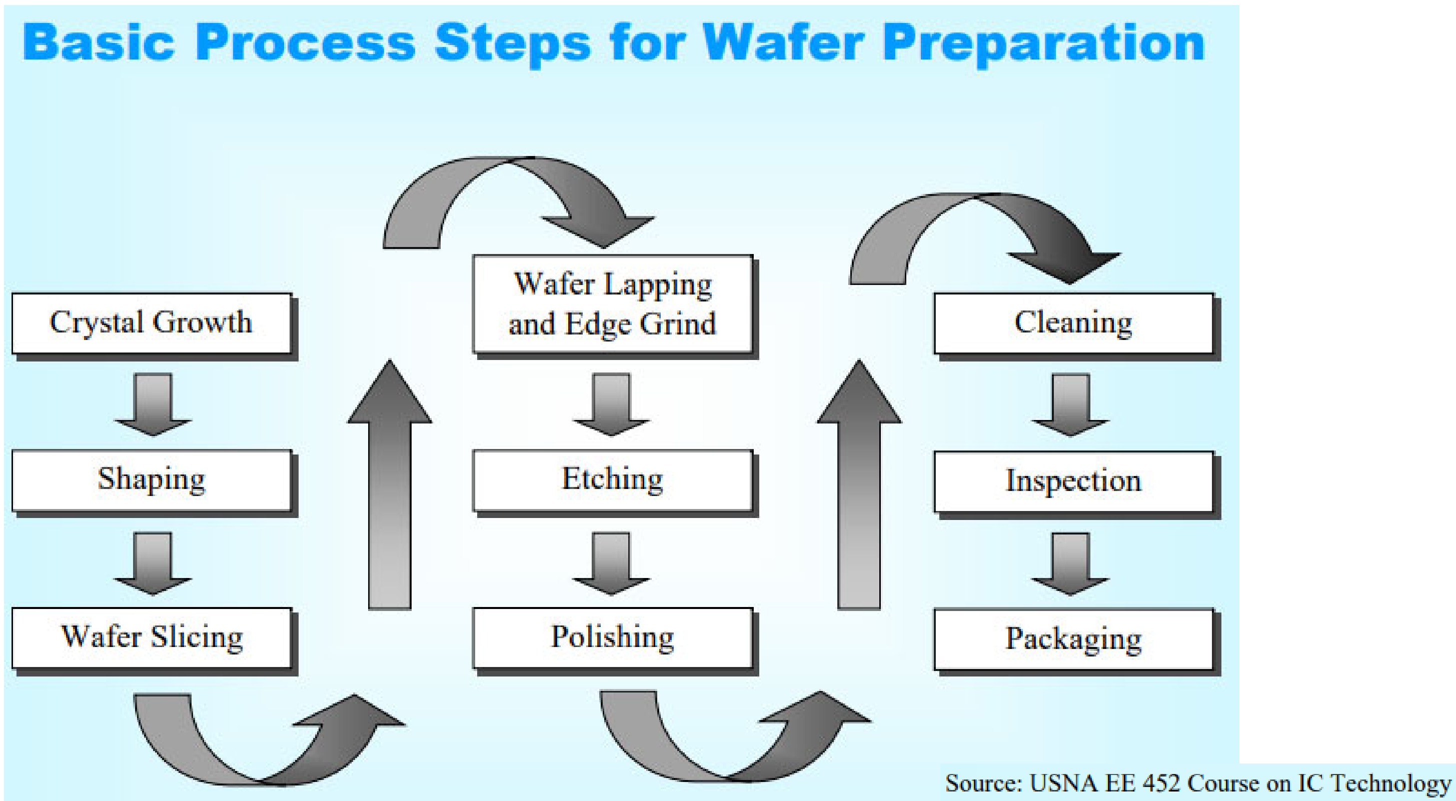
Silicon Wafer Preparation



https://www.physics.muni.cz/~jancely/teamCMV/Texty/RuzneTexty/Krystaly/CrystalStructure&Growth.pdf

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CMOS Fabrication Steps

- Trench isolation to define active regions
- N well definition for PMOS transistors
- P well definition for NMOS transistors
- Gate oxide formation and gate poly silicon patterning
- NMOS halo and shallow extension implant
- PMOS halo and shallow extension implant
- Spacer formation
- N+ select for NMOS source/drain
- P+ select for PMOS source/drain
- S/D activation and Silicidation
- Contact definition
- Metal 1 layer patterning

Ref: N. Bhat, CeNSE, IISc Bangalore- Nanoelectronic Device Technology

CMOS Fabrication Steps

- **CMOS transistors are fabricated on silicon wafers**
- **Lithography process has been the mainstream chip manufacturing process**
- **On each step, different materials are deposited or etched**
- **Easiest to understand by viewing both top and cross-section of wafer in a simplified manufacturing process**

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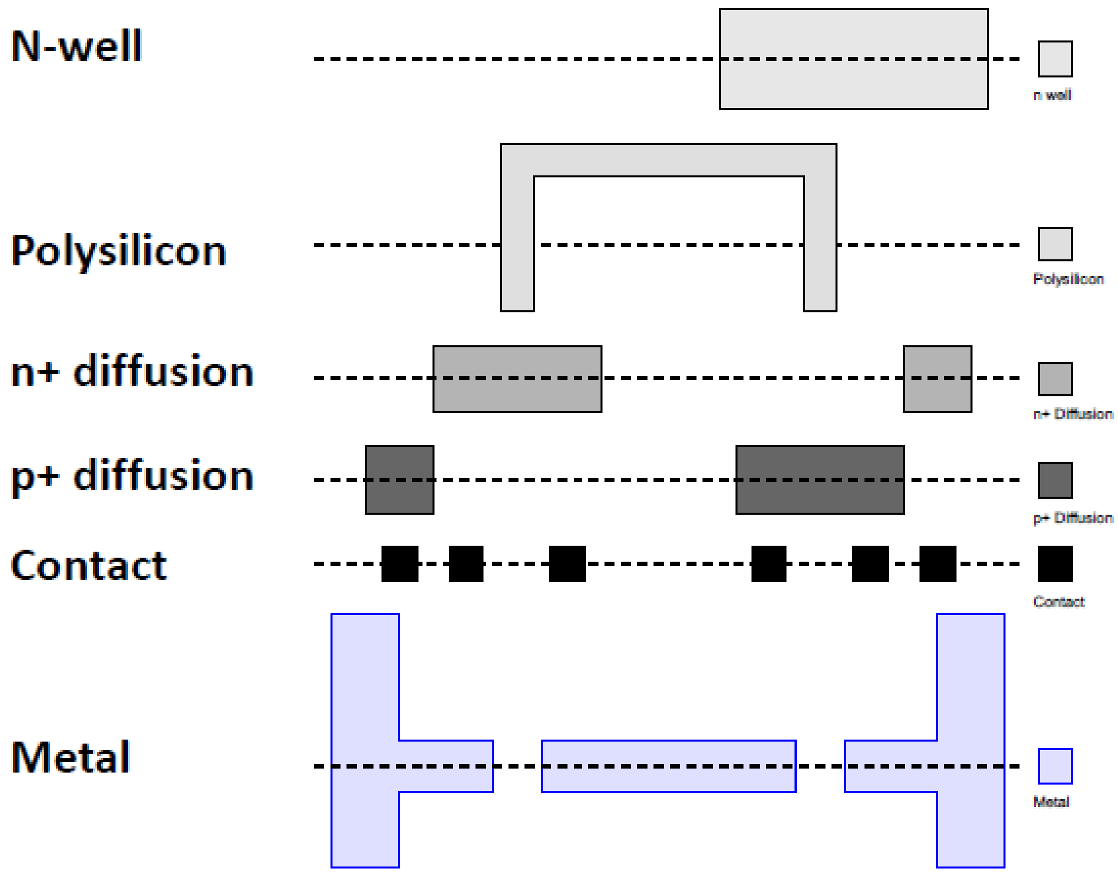
CMOS Fabrication Steps

Six masks to build simple inverter



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IC manufacturing- FEOL and BEOL

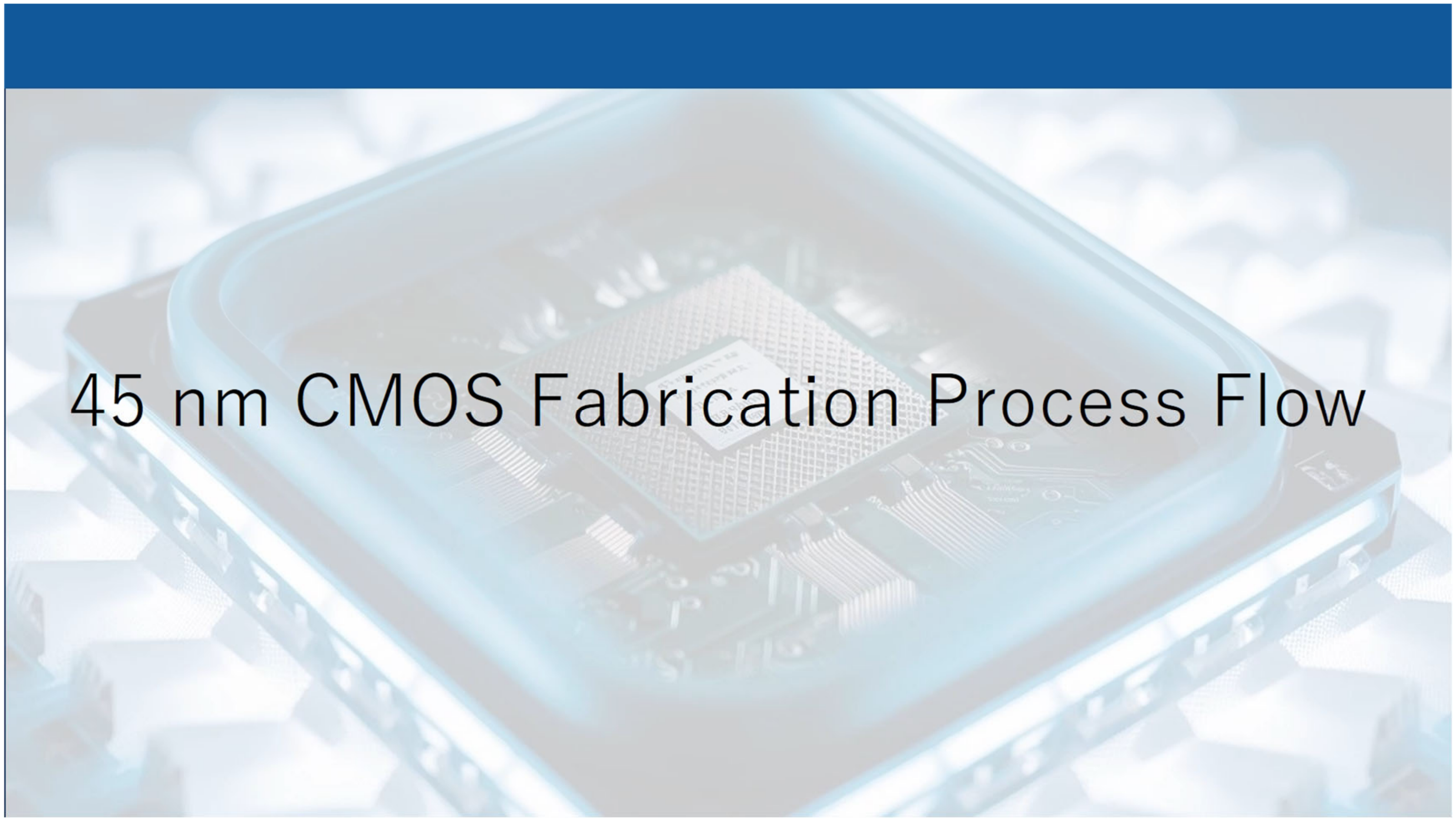
- **FEOL Processes**
  - Si epitaxy
  - Ion implant / diffusion
  - Oxidation
  - Photolithography /Etch
  - RTO

Create the transistors

- **BEOL Processes**
  – PVD/sputtering
  – CVD
  – PECVD
  – Plasma etching
  – Photolithography
  – CMP
  – Electroplating

Do the wiring

MOSFET Fabrication-45nm Technology Node



REF: K. Minakata, Alps Alpine Co., Ltd

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